

Unit 2, Lesson 6: Plotting, Ordering, and Comparing Real Numbers



Benchmark

MA.8.NSO.1.2 Plot, order, and compare rational and irrational numbers, represented in various forms.



Learning Targets

- ☐ I can compare rational and irrational values.
- ☐ I can order rational and irrational values.
- ☐ I can plot rational and irrational numbers on a number line.

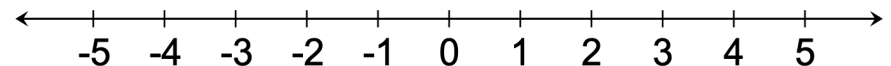


2.6.1 Warm-Up: Bell Work

1. Six real numbers are shown.

$$\sqrt{6}, \quad 1.2, \quad -3\frac{1}{2}, \quad \frac{4}{3}, \quad -\pi, \quad \sqrt[3]{8}$$

- a. Circle all numbers that are irrational.
- b. Order all the values from greatest to least.
- c. Plot and label each number on the number line.





2.6.2 Activity: Station 1 – Comparing and Ordering

1. Compare each pair of expressions using the symbols $<$, $>$, or $=$.

a. -5π $\begin{array}{|c|} \hline \circ < \\ \circ > \\ \circ = \\ \hline \end{array}$ $-\sqrt{225}$

b. $\sqrt[3]{125}$ $\begin{array}{|c|} \hline \circ < \\ \circ > \\ \circ = \\ \hline \end{array}$ $\frac{\sqrt{100}}{2}$

c. $\sqrt{45}$ $\begin{array}{|c|} \hline \circ < \\ \circ > \\ \circ = \\ \hline \end{array}$ $\pi + 4$

d. -7 $\begin{array}{|c|} \hline \circ < \\ \circ > \\ \circ = \\ \hline \end{array}$ $-\sqrt[3]{49}$

e. $\frac{29}{3}$ $\begin{array}{|c|} \hline \circ < \\ \circ > \\ \circ = \\ \hline \end{array}$ $\sqrt{80}$

f. $\frac{\pi}{3}$ $\begin{array}{|c|} \hline \circ < \\ \circ > \\ \circ = \\ \hline \end{array}$ $\sqrt{22} - 3$

2. Choose one pair of expressions from the previous question that were not equivalent.

- a. Circle the pair of expressions.
- b. Explain how you determined which value was greater.

3. Order each set of numbers from least to greatest.

a. $\sqrt{34}$, $\pi + 2$, $5\frac{1}{2}$

b. $-\sqrt[3]{60}$, $-\frac{18}{4}$, $-\pi$



2.6.3 Activity: Station 2 – Plotting Rational and Irrational Numbers on a Number Line

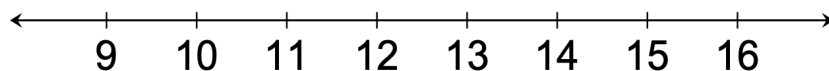
1. Five expressions are shown.

$$\sqrt{150} \quad -\pi + 15 \quad 10.2 \quad 11\frac{1}{3} \quad \sqrt[3]{64} + 8$$

- a. Write each expression under the appropriate description in the table.

Includes Only Rational Numbers	Includes an Irrational Number

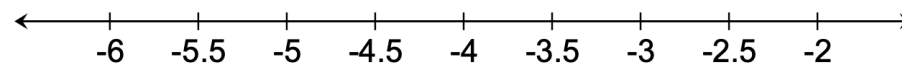
- b. Plot and label each number on the number line.



2. Six expressions are shown.

$$-\sqrt{20} \quad -\frac{60}{11} \quad -\sqrt{8} \quad -\pi \quad \sqrt[3]{-125} \quad -\frac{23}{5}$$

Plot and label the location of each expression on the number line.

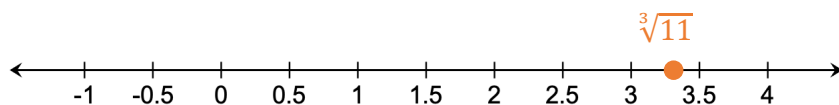




2.6.4 Activity: Station 3 – Error Analysis

1. Sara approximated the value of $\sqrt[3]{11}$. Her thinking is shown.

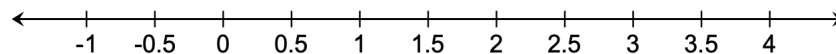
The number 11 is between 9 and 16, which are both perfect squares. So, I know $\sqrt[3]{11}$ lies between 3 and 4 on a number line, but it's closer to 3 because 11 is closer to 9 than 16. I would plot its approximate location here.



- a. Discuss Sara's thinking with your partner. Then, complete the table.

What is correct in Sara's thinking?	What error(s) did Sara make?

- b. Plot and label the correct approximate value of $\sqrt[3]{11}$ on the number line.



2. For each pair of numbers, an explanation comparing the values is provided.

- a. Determine whether each explanation is correct or incorrect.

Pair of Numbers	Comparison and Explanation	Is it Correct?
π^2 and 9	9 is greater than π^2 because π is slightly more than 3, so π^2 is slightly more than 6.	☐ correct ☐ incorrect
$\sqrt{50}$ and 8	$\sqrt{50}$ is less than 8 because 8 is equal to $\sqrt{64}$ and $\sqrt{50}$ is less than that.	☐ correct ☐ incorrect
$\frac{13}{3}$ and $\sqrt[3]{60}$	$\frac{13}{3}$ is greater than $\sqrt[3]{60}$ because $\frac{13}{3}$ is slightly more than 4 and $\sqrt[3]{60}$ is slightly less than 4.	☐ correct ☐ incorrect
-2π and -6	-6 is less than -2π because $-2 \cdot \pi$ is between -6 and -7 , which means it's larger than -6 .	☐ correct ☐ incorrect

- b. Two of the explanations above are incorrect. Write the correct explanation for each of them by completing the sentence stems.

_____ is greater than _____ because ...

_____ is less than _____ because ...



2.6.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept learned in this unit.



I need help.



I got it.



I could help someone.

Solve equations in the form $x^2 = p$.	
Solve equations in the form $x^3 = q$.	
Calculate square roots of perfect squares and cube roots of perfect cubes.	
Identify whether a number is rational or irrational.	
Locate the approximate value of expressions involving irrational numbers on a number line.	
Plot, order, and compare rational and irrational numbers on a number line.	